Tutorial 1

EC3301 - Martinmas Semester

Question 1

In each case, try to prove the following properties of the OLS estimators $(\hat{\beta}_0, \hat{\beta}_1)$ for a simple linear regression model

$$y_i = \beta_0 + \beta_1 x_i + u_i$$

a. $\sum_{i=1}^{n} \hat{u}_i = 0$, where $\hat{u}_i = y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i$

b. $\bar{y} = \hat{\beta}_0 + \hat{\beta}_1 \bar{x}$, where $\bar{y} = \frac{1}{n} \sum_{i=1}^{n} y_i$ and $\bar{x} = \frac{1}{n} \sum_{i=1}^{n} x_i$.

c. $\sum_{i=1}^{n} x_i \hat{u}_i = 0$

d. $\widehat{Cov}(x_i, \hat{u}_i) = 0$ where $\widehat{Cov}()$ is the sample covariance

e. $\widehat{Cov(y_i}, \hat{y}_i) = \widehat{Var}(x_i)$ where $\widehat{Var}()$ is the sample variance

Maybe add page 26 - covariance formula.

Question 2

Need some hypothesis tests

Question 3